VANISHING OF THE BOCKSTEIN BETA VERSUS LIFTING KERNELS MODULO F IN TERMS OF F^2

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Allowed standard prerequisites (editorial).

- The boundary map beta on the cohomology of the quotient complexes fM/f^2M, M/f^2M, M/fM computes the obstruction to lifting kernels of d mod f to kernels of d mod f^2. invoked at proof lines 27797-27801. Gives the equivalence of (1) and (2).
- The Bockstein beta factors through a map whose target detects the f-torsion of H^{i+1}(M). invoked at proof lines 27802-27803. Shows nonzero beta forces H^{i+1}(M)[f] nonzero.

Source reference. Original equation 0GSP: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27671–27671. Cited at proof line 27803 to force f-torsion from a nonzero beta; the factorization is used inside the counted span.

Lemma 0.2. *Let A be a ring and let $f \in A$ be a nonzerodivisor. Let $M^\bullet$ be a complex of f-torsion free A-modules. For $i \in \mathbf{Z}$ the following are equivalent*

(1) $\mathrm{Ker}(d^i \bmod f^2)$ *surjects onto* $\mathrm{Ker}(d^i \bmod f)$,
(2) $\beta : H^i(M^\bullet \otimes_A f^i A/f^{i+1}A) \to H^{i+1}(M^\bullet \otimes_A f^{i+1}A/f^{i+2}A)$ *is zero.*

These equivalent conditions are implied by the condition $H^{i+1}(M^\bullet)[f] = 0$.

Proof. The equivalence of (1) and (2) follows from the definition of β as the boundary map on cohomology of a short exact sequence of complexes isomorphic to the short exact sequence of complexes $0 \to fM^\bullet/f^2M^\bullet \to M^\bullet/f^2M^\bullet \to M^\bullet/fM^\bullet \to 0$. If $\beta \neq 0$, then $H^{i+1}(M^\bullet)[f] \neq 0$ because of the factorization (0GSP). □

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